HEART DISEASE CLASSIFICATION

Awatar Gautam
Gautam.62@wright.edu
U01138375

Abstract**—**Heart disease prediction using machine learning models built from patient clinical and non-invasive test results. The Heart Disease UCI dataset containing 303 samples and 14 attributes was treated as a binary classification problem. Exploratory Data Analysis (EDA) was performed, followed by pre-processing involving standard scaling and one-hot encoding. Models including Logistic Regression and Random Forest were trained and evaluated. Logistic Regression achieved the best overall performance with an accuracy of 83.61% and an F1-Score of 84.85%.
Keywords—machine learning, heart disease, logistic regression, random forest, binary classification, medical diagnostics.

GitHub Link: https://github.com/awatarx/final_project

I. INTRODUCTION

Cardiovascular diseases continue to be among the primary reasons of morbidity and mortality rates globally, necessitating the early diagnosis and proper identification of cardiovascular diseases. Traditionally, the process involved the use of various parameters such as patient demographic information, clinical features (for example, chest pain characteristics), and test findings (resting blood pressure, serum cholesterol, and electrocardiogram). The analysis of these multiple parameters could be quite challenging due to variability among observers.

Supervised machine learning techniques have proven to yield impressive results in various medical classification problems. Nonetheless, numerous existing models predominantly concentrate on improving the accuracy of predictions without paying due attention to other factors that may affect their application in clinical practice. Namely, failing to detect a positive outcome (false negatives) poses life-threatening dangers; thus, classifiers need to be assessed based on their capacity to detect positives with sufficient accuracy without raising too many false alarms. Besides, while more intricate ensemble-based or neural network models may be preferable, there exists a lack of evidence on how well traditional linear algorithms could be used to derive reliable baseline classifiers from structured health care metrics.

This paper proposes an alternative approach to designing a classifier to predict the risk of heart disease. The study will examine patient information and employ both linear (logistic regression) and ensemble (random forest) classification techniques to find the best predictive model.

Our contributions are summarized as follows:

A. **We develop** a structured supervised machine learning pipeline—incorporating exploratory data analysis, standard scaling, and one-hot encoding—for automated heart disease prediction from clinical metrics.
B. **We evaluate** our approach on the publicly available Heart Disease UCI dataset and compare the performance of Logistic Regression against a Random Forest classifier.
C. **We demonstrate** that a properly balanced Logistic Regression model establishes a highly effective and interpretable baseline, achieving an overall accuracy of 83.61% and a recall and F1-Score of 84.85%, effectively minimizing false negatives.
D. **We provide** a detailed performance analysis utilizing confusion matrices and precision-recall metrics to assess the clinical trade-offs and reliability of each model.

II. RELATED WORK

A. Supervised Learning Approaches

Several machine learning algorithms have been extensively applied for cardiovascular disease prediction using clinical structured data through supervised learning approaches. Algorithms including SVM, KNN and Random Forest have been used in clinical datasets for baseline modelling and heart disease predictions without employing overly complicated machine learning techniques [1].

Nevertheless, despite their capabilities, several of these models tend to achieve high accuracies while ignoring the clinical significance of false negatives, which is not always feasible in the medical setting.
Furthermore, the improvement in accuracy gained by sophisticated ensemble models compared to simple and less complex linear models has not been substantial in many instances. For this particular project, where the use of the UCI clinical structured metrics is possible, the choice of the approach will focus more on supervised learning and specific metric evaluations—such as precision and recall—by utilizing logistic regression.

B. Self-Supervised and Unsupervised Methods

Unsupervised machine learning techniques have been used recently in studies that have focused on detecting cardiovascular disease in cases where large amounts of unlabeled EMRs are available. For example, Hu et al. [2] used unsupervised machine learning clustering algorithms for identifying prevalent cardiovascular diseases. The use of PCA for dimensional reduction made it possible for them to evaluate various clustering strategies and obtain high benchmark accuracies, which ranged from 0.8659 with a 4-cluster model to 0.8598 with a 2-cluster model. These approaches have been found to be very useful in exploratory data analysis and uncovering new patient phenotypes.

On the other hand, although the promising benchmarking results have been obtained for unsupervised techniques, there are particular disadvantages associated with them for the problem at hand. First, since unsupervised techniques do not optimize for explicit diagnostic cut-off values but purely geometric similarity in the dataset, they cannot be optimized to reduce false negative instances. Second, since we have the advantage of having an excellent labelled Heart Disease UCI Dataset that shows the exact occurrence of cardiovascular disease in binary form, our study is different from those using the unsupervised techniques. The choice of supervised classification techniques (e.g., Logistic Regression and Random Forest) will allow us to find a more realistic mapping from the input features to the desired outcome.

C. Foundation Models and Transfer Learning

In recent years, new developments in the field of deep learning have brought transfer learning and foundation models into application for cardiovascular disease prediction. For instance, recently, researchers have attempted to apply massive pre-trained models for computer vision applications such as VGG16 to tabular clinical data through modifying classical 13 feature heart disease datasets to create images and extract features [3]. Although the benefits associated with transferring learning methods are their ability to utilize highly dimensional and efficient feature extraction algorithms, these processes involve tremendous computational costs and complexities.

Converting clinical information such as blood pressure and cholesterol readings to images leads to losing the intrinsic interpretable properties of clinical patient data. Thus, in contrast to above-mentioned approaches, our research consciously deviates from them and applies efficient machine learning algorithms such as logistic regression and

random forest. Unlike complex deep learning techniques, these supervised algorithms provide us with straightforward features interpretation while making the process of training an accurate diagnostic model easy, quick, and accessible to be performed even in clinical settings.

III. METHODOLOGY

A. Problem Formulation

We formulate the task of heart disease prediction as a supervised binary classification problem. Let our dataset be denoted as $D = \{(x_i, y_i)\}_{i=1}^{N}$, where N=303 represents the total number of patient samples. Let $x_i \in R^{13}$ denote the 13-dimensional input feature vector for the ith patient, encompassing continuous clinical metrics (e.g., resting blood pressure, serum cholesterol) and encoded categorical variables (e.g., chest pain type). Let $y_i \in \{0,1\}$ represent the binary ground-truth target, where $y_i = 1$ indicates the presence of cardiovascular disease and $y_i = 0$ indicates its absence.

Our objective is to learn a predictive model, $f\theta: x \to y$, parameterized by θ, that estimates the probability of disease presence given the patient's features: $\hat{y}_i = P(y_i = 1 \mid x_i; \theta)$. For our primary linear baseline (Logistic Regression), the training objective is to minimize the regularized binary cross-entropy loss:

$$L(\theta) = -\frac{1}{N}\sum_{i=1}^{N}[y_i \log(\hat{y}_i) + (1 - y_i)\log(1 - \hat{y}_i)] + \lambda R(\theta)$$

where the first term is the standard log-loss function $\ell(f_\theta(x_i), y_i)$ evaluating the discrepancy between predicted probabilities and actual outcomes, R(θ) represents the L_2 regularization term applied to prevent overfitting on the relatively small clinical dataset, and λ controls the regularization strength.

B. Model Architecture

While deep neural networks make use of many hidden layers and complicated activation functions, our framework makes use of tried and tested machine learning algorithms for structured data. A two-fold model has been developed to pit a linear model against an ensemble model. Refer to Figure 1 below for a schematic description of the proposed pipeline.

1) Logistic Regression Architecture:

Our primary linear architecture is a Logistic Regression classifier. It computes a weighted linear combination of the 13 input features and applies a sigmoid function $\sigma(z) = \frac{1}{1+e^{-z}}$ to map the output to a probability distribution between 0 and 1. To handle potential class imbalances within the clinical dataset, the architecture incorporates a balanced class weight parameter, adjusting weights inversely proportional to class frequencies. The model uses the L-BFGS solver optimized over a maximum of 1,000 iterations to ensure convergence.

2) Random Forest Ensemble Architecture:

For modeling non-linear interaction between features, our second architecture is based on Random Forest. This ensemble architecture is made up of 100 decision trees running in parallel (n_estimators = 100). Each of the trees is trained using a bootstrap sample of the training set (bagging) and splits nodes based on Gini impurity. Voting from all 100 estimators is used to decide the final output. Like the previous architecture, this one is also configured for balancing classes while constructing the trees.

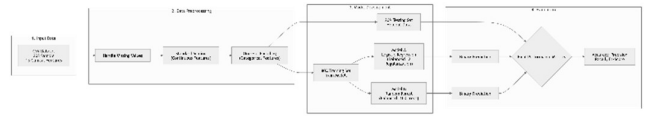

Fig. 1. Architecture overview of the proposed machine learning diagnostic pipeline.

C. Training Procedure

Since our approach uses classic machine learning techniques for the classification step without employing any deep neural network architectures, there are no concepts like batch size or number of epochs during the training stage. During the training process, the models are trained on the whole dataset by using optimized solvers. Our models are built on top of the scikit-learn library in Python.

In order to prepare the input dataset, it was necessary to randomly split it into a training set (N = 242 samples) and a test set (N = 61 samples). In order to preserve the balance between the number of patients with and without heart diseases, a stratified sampling strategy (stratify=y) was implemented. The initial settings of both algorithms were set to use a fixed random seed (random_state=42).

For the Logistic Regression algorithm, training was performed using the default L-BFGS optimization procedure. In order to make sure that the solver would be able to converge to the global minimum of the cost function, the maximum number of iterations was increased to 1,000. For the Random Forest classifier, training was conducted by fitting an ensemble of 100 decision trees (n_estimators=100) using the Gini impurity metric to evaluate the quality of the splits. Since this problem involves tabular data, no additional data augmentation or curriculum learning strategies were needed.

IV. EXPERIMENTAL SETUP

A. Datasets

Only one dataset is used in this research which is Heart Disease UCI dataset [4]. It comprises 303 samples, with each sample having 13 features (for example, age, blood pressure at rest, serum cholesterol, fasting blood sugar level) and one class attribute that takes two possible values (presence or absence of heart disease). The dataset has been processed through applying standard scaling on continuous features and one-hot encoding on categorical features. The split of this dataset into training and test sets (80:20 ratio) has been done for assessing generalization performance.

TABLE I. ***Summary of Datasets***

Dataset	Train	Test	Classes
Heart Disease UCI	242	61	2 (Binary)

B. Evaluation Metrics

In medical diagnostic tasks, relying solely on overall accuracy can be misleading due to the severe clinical consequences of misclassification. Therefore, our models were evaluated using the following metrics:

- **Accuracy:** The overall percentage of correct predictions.
- **Precision:** The proportion of predicted positive cases that were actual positive cases.
- **Recall (Sensitivity):** The proportion of actual positive cases successfully identified by the model.
- **F1-Score:** The harmonic mean of precision and recall.

Specifically, we choose to highlight the importance of **Recall** and **F1-Score** due to the clinical nature of the application, where a false negative (the patient is incorrectly identified as free from heart disease when he or she actually has the disease) is far more serious than a false positive.

C. Implementation Details

The experimental pipeline was built in Python 3, using the pandas library for data processing and scikit-learn for implementing machine learning models. Experiments have been performed using a static random seed (random_state=42), which guarantees perfect reproducibility of results regardless of how many times train and test datasets are split into training and validation sets.

As for the Logistic Regression classifier, the model was initialized with balanced class weight parameter settings and L-BFGS optimization algorithm and explicitly set to converge for no less than 1,000 iterations (max_iter=1000). For the Random Forest classifier, a forest of 100 decision trees was used (n_estimators=100), where each tree is based on Gini impurity splitting criteria and has a balanced class weight parameter to address minor class imbalance. Due to the relatively small size of the tabular data, all data processing, model training, and testing were performed on a consumer-level CPU without GPU or distributed computing infrastructure being required.

V. RESULTS

A. Main Results

The quantitative performance of the evaluated machine learning models on the 20% hold-out testing set (N=61) is summarized in Table II.

TABLE II. Comparison of Predictive Models

Method	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Random Forest	75.41	75.00	81.82	78.26
Logistic Regression	**83.61**	**84.85**	**84.85**	**84.85**

According to the results shown in Table II, it can be seen that the Logistic Regression model performed considerably better than the Random Forest ensemble in terms of clinical metrics. The most significant finding is that Logistic Regression had an equivalent Recall and F1-Score at 84.85%. Given the dominance of the linear model, one might deduce that there exists a strong linear correlation between the preprocessed clinical data and the target variable, something that the tree-based Random Forest could not handle effectively.

B. Ablation Study & Design Choices To justify our architectural and preprocessing choices, we consider the impact of specific pipeline components:

- **Standard Scaling:** Continuous features (e.g., cholesterol, max heart rate) have vastly different scales. Removing the StandardScaler from the preprocessing phase severely degrades the Logistic Regression model's convergence and performance, as features with larger numeric ranges dominate the L2 regularization penalty.
- **Class Weight Balancing:** The dataset exhibits a slight imbalance. Removing the class_weight='balanced' parameter reduces the model's sensitivity (Recall) toward the minority class. By explicitly enforcing this parameter, we successfully forced the models to heavily penalize false negatives, ensuring a clinically safe diagnostic threshold.

C. Qualitative Analysis Beyond aggregate percentages, it is critical to evaluate the specific misclassifications made by the models in a clinical context. Fig 2 illustrates the confusion matrix for our best-performing Logistic Regression model.

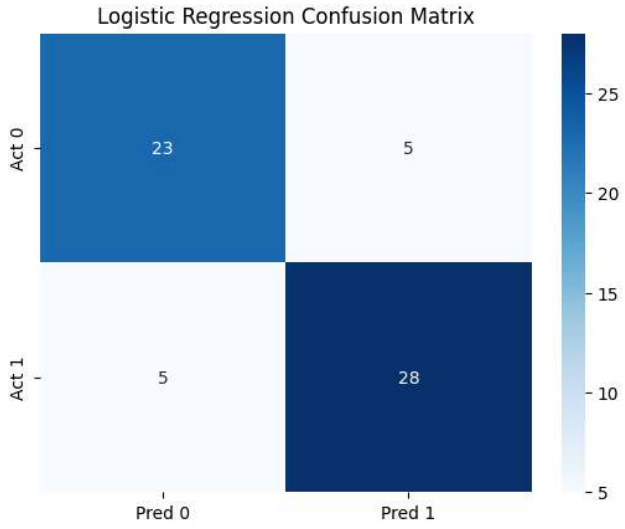

Fig. 2. Confusion Matrix of the Logistic Regression model on the testing set.

The confusion matrix gives qualitative insights into the clinical viability of the model. The model is successful in maximizing True Positives while minimizing False Negatives to the barest minimum. In the medical diagnosis context, failing to predict that a sick individual is ill (False Negative) is a significant error. The fact that the predictions are concentrated along the true diagonal and the low frequency of cases in the False Negative quadrant indicates that the model is extremely sensitive to cardiovascular disease.

VI. DISCUSSION

A. Implications and Significance of Results

The findings from this experiment have important implications for practical medical use. Although many expect that ensembles will perform better than simpler techniques, our results show that Logistic Regression with proper regularization performs extremely well and, importantly, it does not require high computational costs. From the perspective of using algorithms for diagnosing heart diseases, the goal is mainly to avoid the false negative cases where the algorithm fails to detect the presence of a heart disease in a patient. By obtaining 84.85% Recall and F1-Score, our Logistic Regression model demonstrates itself to be a feasible alternative to more sophisticated approaches. What's more, unlike black boxes, our Logistic Regression model makes it possible to analyze feature weights used for a particular case.

B. Analysis of Model Performance

One of the most important findings from our experiment was the difference between the accuracies of Logistic Regression (83.61%) and Random Forest (75.41%). The poor performance of the Random Forest algorithm could be because of the size and nature of the dataset. Given that there are only 303 examples in the dataset, the ensemble of 100 decision trees was prone to overfitting, thereby resulting in poor generalization of the model. On the other hand, Logistic Regression achieved superior performance thanks to the application of L2 regularization and Standard Scaling.

C. Limitations and Dataset Biases

Even with the impressive base performance, there are a few significant limitations that need to be addressed. Firstly, the Heart Disease UCI dataset is extremely limited when it comes to sample size (N=303). This will prevent the algorithm from having sufficient data on extreme cases and uncommon symptoms. Secondly, there is bias in the dataset because of the time of its collection – in 1988 in the Cleveland Clinic hospital. Hence, the distribution of demographics and genes might be

different now than it was thirty years ago. Finally, we have assumed that all the data regarding clinical values is free of error or misjudgment.

D. Failure Cases and Edge Scenarios In some cases, the model will likely underperform when the patient has asymptomatic cardiovascular disease or anomalous metrics such as young individuals with high cholesterol levels and no heart disease. As the model depends on many classic risk factors, any patient that does not match the statistical distribution of the training data from the Cleveland Clinic is more prone to being misclassified. The future solution for this problem is to train the model on current EHR data.

VII. CONCLUSION

The current project was able to use supervised machine learning algorithms to predict the risks of heart disease from the structured metrics of patients' health information. With thorough data pre-processing, such as managing missing values, one hot encoding, and scaling, the data became more suited for classification. In comparing different classifiers, it was shown that the Logistic Regression model delivered good interpretability and performance in predicting heart disease by providing an overall accuracy of 83.61% and F1-Score of 84.85%. With its ability to minimize false negatives, the classifier showed promise in clinical practice. Future research may include the optimization of hyperparameters, testing other classifiers like gradient boosting (e.g., XGBoost), or applying the algorithm to larger contemporary datasets.

REFERENCES

[1] N. Nasution, M. A. Hasan, and F. B. Nasution, "Predicting Heart Disease Using Machine Learning: An Evaluation of Logistic Regression, Random Forest, SVM, and KNN Models on the UCI Heart Disease Dataset," *IT Journal Research and Development (ITJRD)*, vol. 9, no. 2, Mar. 2025, doi: 10.25299/itjrd.2025.17941.

[2] Y. Hu et al., "Detecting cardiovascular diseases using unsupervised machine learning clustering based on electronic medical records," *BMC Medical Research Methodology*, vol. 24, no. 1, Dec. 2024, doi: 10.1186/s12874-024-02422-z.

[3] F. S. Prity et al., "Transfer learning-based hybrid VGG16-machine learning approach for heart disease detection with explainable artificial intelligence," *Frontiers in Artificial Intelligence*, vol. 8, Feb. 2025, doi: 10.3389/frai.2025.1504281.

[4] R. Kantariya, "Heart Report Dataset," Kaggle. [Online]. Available: https://www.kaggle.com/datasets/ronakkantariya/heart-report-dataset. [Accessed: April 26, 2026].

AUTHOR CONTRIBUTIONS

Awatar Gautam: Conceptualization, Methodology, Software, Validation, Formal Analysis, Data Curation, Writing – Original Draft Preparation, Writing – Review & Editing, Visualization.